

```

#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>

int main()
{
    char cmd[100];

    printf("Enter Linux command: ");
    scanf("%s", cmd);
    pid_t pid = fork();
    if(pid == 0)
    {
        printf("Child PID : %d\n", getpid());
        execlp(cmd, cmd, NULL);
        perror("Execution Failed");
    }
    else if(pid > 0)
    {
        printf("Parent PID : %d\n", getpid());
        wait(NULL);
        printf("Child process completed.\n");
    }
    else
    {
        printf("Fork Failed\n");
    }
    return 0;
}

```

Try uname, lscpu, lsblk, ps, top commands after this program

```
GNU nano 8.7.1 nikhil.c
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>

int main()
{
    char cmd[100];

    printf("Enter Linux command: ");
    scanf("%s", cmd);

    pid_t pid = fork();

    if (pid == 0)
    {
        printf("Child PID : %d\n", getpid());
        execlp(cmd, cmd, NULL);

        perror("Execution Failed");
    }
    else if (pid > 0)
    {
        printf("Parent PID : %d\n", getpid());

        wait(NULL);

        printf("Child process completed.\n");
    }
    else
    {
        printf("Fork Failed\n");
    }

    return 0;
}
```

```
nikhilreddy@Hp: ~$ ls
example.txt  filename.c  nikhil  nikhil.c  nikhill  nikhill.c  nikhilprograms  nikhilreddy
nikhilreddy@Hp: ~$ nikhil.c
nikhil.c: command not found
nikhilreddy@Hp: ~$ nano nikhil.c
nikhilreddy@Hp: ~$ ls
example.txt  filename.c  nikhil  nikhil.c  nikhill  nikhill.c  nikhilprograms  nikhilreddy
nikhilreddy@Hp: ~$ gcc program.c -o nikhil
ccl: fatal error: program.c: No such file or directory
compilation terminated.
nikhilreddy@Hp: ~$ gcc nikhil.c -onikhil
nikhilreddy@Hp: ~$ ls
example.txt  filename.c  nikhil  nikhil.c  nikhill  nikhill.c  nikhilprograms  nikhilreddy
nikhilreddy@Hp: ~$ gcc nikhil.c -onikhil
nikhilreddy@Hp: ~$ ./nikhil
Enter Linux command: ls
Parent PID : 625
Child PID : 626
example.txt  filename.c  nikhil  nikhil.c  nikhill  nikhill.c  nikhilprograms  nikhilreddy
Child process completed.
nikhilreddy@Hp: ~$
```

2. copying the contents from one file to another file

```

#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>

int main()
{
    int src, dest;
    char buffer[1024];
    int bytes;

    src = open("input.txt", O_RDONLY);

    if(src < 0)
    {
        printf("Cannot open source file\n");
        return 1;
    }

    dest = open("output.txt", O_WRONLY | O_CREAT | O_TRUNC, 0644);

    while((bytes = read(src, buffer, sizeof(buffer))) > 0)
    {
        write(dest, buffer, bytes);
    }

    close(src);
    close(dest);

    printf("File copied successfully.\n");

    return 0;
}

```

```

GNU nano 8.7.1 nikhil.c
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>

int main()
{
    int src, dest;
    char buffer[1024];
    int bytes;

    src = open("input.txt", O_RDONLY);

    if (src < 0)
    {
        printf("Cannot open source file\n");
        return 1;
    }

    dest = open("output.txt", O_WRONLY | O_CREAT | O_TRUNC, 0644);

    while ((bytes = read(src, buffer, sizeof(buffer))) > 0)
    {
        write(dest, buffer, bytes);
    }

    close(src);
    close(dest);

    printf("File copied successfully.\n");

    return 0;
}

```

^G Help ^O Write Out ^F Where Is ^X Cut [Read 32 lines] ^G Location ^U Undo ^-A Set Mark ^-] To Bracket
 ^X Exit ^R Read File ^A Replace ^U Paste ^! Execute ^_ Go To Line ^-E Redo ^-C Copy ^_ Where Was

```
nikhilreddy@Hp:~$ gcc nikhill.c -onikhil
nikhilreddy@Hp:~$ ./nikhil
File copied successfully.
nikhilreddy@Hp:~$ nano input.txt
gcc nikhill.c -onikhil
./nikhil
cat output.txt
File copied successfully.
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>

int main()
{
    int src, dest;
    char buffer[1024];
    int bytes;

    src = open("input.txt", O_RDONLY);

    if (src < 0)
    {
        printf("Cannot open source file\n");
        return 1;
    }

    dest = open("output.txt", O_WRONLY | O_CREAT | O_TRUNC, 0644);

    while ((bytes = read(src, buffer, sizeof(buffer))) > 0)
    {
        write(dest, buffer, bytes);
    }

    close(src);
    close(dest);

    printf("File copied successfully.\n");

    return 0;
}
nikhilreddy@Hp:~$ |
```